Arwin Karir

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Education

University of Toronto Sept. 2025 – May 2030
BASc in Computer Engineering (PEY Co-op) *Toronto, ON*

Experience

Software Engineering Intern May 2026 – Present
SellStatic *Toronto, ON*

- Implementing feedback automation architecture with AWS SES, scheduled cron jobs, tokenized feedback links, unsubscribe handling, and database-backed email state to collect post-signup customer experience feedback
- Designing and developing an AI-native browser-based video editor by improving React/Next.js editor workflows for media upload/loading, timeline editing, text overlays, animations, preview playback, autosave, and export across a complex interactive UI
- Spearheading end-to-end development of a website chatbot feature by architecting the chat UI, integrating a GPT API plugin, implementing request/response handling, and designing production-ready interaction flows for user support and engagement
- Debugging and improving project persistence by tracing reload-related state loss across frontend state, Prisma/PostgreSQL save/load flows, and persistent media storage, helping restore videos, timeline clips, text layers, and draft metadata after refresh
- Developing QA workflows with Playwright, Playwright MCP, network/console monitoring, cross-browser testing, and upload/playback/export edge cases to validate editor reliability before release

Artificial Intelligence Intern Jun. 2023 – Aug. 2023
Amazon Work Experience Program *Remote*

- Built a supervised machine learning workflow on environmental datasets for forest-cover classification, structuring experiments for reproducible training, evaluation, and model comparison
- Owned Python-based preprocessing and feature engineering with pandas, NumPy, and scikit-learn to clean inputs, standardize features, and improve training signal quality
- Benchmarked k-NN, logistic regression, and neural network models, reporting trade-offs across accuracy, interpretability, and computational cost to guide model selection
- Used AWS development workflows with Amazon SageMaker, Amazon S3, and Amazon EC2, collaborating on a 4-person team to present metrics-driven results and next steps to AWS executives

Projects

LumiSense AI | *PyTorch, ONNX, OpenCV, Raspberry Pi* Jan. 2026 – Apr. 2026

- Implemented a stakeholder-facing edge AI solution for the UofT Sustainability Office to estimate floor-plane illuminance and support campus building lighting performance analysis
- Developed a multitask vision model to estimate floor-plane illuminance by predicting a dense lux map, summary statistics (avg lux, p5, p95), and point-wise lux queries under/between fixtures
- Implemented an encoder–decoder architecture with a ResNet-style backbone, U-Net-style upsampling, and task-specific heads for segmentation, lux regression, and uncertainty/quality estimation
- Exported models to ONNX and deployed an edge inference loop on a Raspberry Pi using onnxruntime and OpenCV camera capture, enforcing preprocessing parity between training and inference
- Added deployment reliability features including luminance-based frame-quality gating, debug logging, and snapshots while generating outputs for UofT Sustainability Office lighting performance reporting

TruthLens | *Python, NLP, TF-IDF, Embeddings* Jan. 2026 – Mar. 2026

- Implemented an end-to-end NLP pipeline that classifies articles as fake (0) vs real (1), structuring the notebook for reproducibility and extension to production workflows
- Applied regex cleaning, stopword removal, and lemmatization while engineering TF-IDF features for lightweight baseline models
- Compared classical ML baselines against transformer-based contextual embeddings, analyzing trade-offs in accuracy, compute cost, and generalization
- Evaluated models using accuracy, precision, recall, F1, and confusion matrices; documented upgrade paths for hyperparameter tuning, ensembles, real-time inference, APIs, and dashboards

Technical Skills

Languages: Python, Swift, C/C++, JavaScript/TypeScript, HTML/CSS, MATLAB, SQL
Web / Full-Stack: Next.js, React, Node.js, Prisma, PostgreSQL, REST APIs,
Testing / Quality: Playwright, Playwright MCP, end-to-end testing, API testing, console/network monitoring, accessibility, performance, security, cross-browser validation
Machine Learning / Data: PyTorch, scikit-learn, pandas, NumPy; transformer embeddings; CNNs; evaluation (F1/precision/recall), confusion matrices
Mobile / CV / Edge: ARKit, SwiftUI, CoreML, Vision; ONNX Runtime, OpenCV; Raspberry Pi inference pipelines
Cloud / DevOps: AWS (SES, S3, SageMaker, EC2), Azure,, scheduled jobs/cron, lifecycle email automation, Git/GitHub, Linux, VS Code, Xcode, Jupyter, Google Colab